Conversion of Camphene to Camphor

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Introduction

In this week's lab, borneol was converted to camphor through reflux. Reflux is the addition of heat, without the evaporation of liquid.

Materials and Methods

First, 0.3 g of the borneol product was added to a long-neck 10 mL round-bottom flask with 0.3 mL of glacial acetic acid. The racemic mixture was dissolved in 1 mL of acetone and a boiling chip was added. Next, a distillation column was assembled for reflux and the apparatus was placed in a water bath with a temperature of around 40-50℃. After the reagents were dissolved, 6 mL of aqueous sodium hypochlorite solution was added dropwise over the span of 35 minutes. The solution was swirled gently after each addition to ensure all the chemicals were mixed thoroughly. After the last addition of bleach, the reaction was left to continue for 15 minutes. Next, the mixture was left to cool to room temperature, and the camphor was extracted with two portions of 2 mL of dichloromethane. The methylene chloride layers were then neutralized using 2 mL of saturated sodium bicarbonate. The organic solution was washed sequentially with 2 mL of 5% sodium bisulfate and 2 mL of water. The organic solution was then dried using anhydrous sodium sulfate. Finally, the dichloromethane was evaporated over a water bath and the crystals were weighed.

Results

Weight of watch glass	29.645 g
Weight of watch glass and product	29.814 g
Weight of product	0.169 g

Table 1. Weight of watch glass, weight of watch glass and product, weight of product

Calculations

Weight of product: 29.814 g - 29.645 g = 0.169 grams Camphor

Percent Yield: actual yield / theoretical yield x 100% = percent yield

Theoretical Yield: 0.3 g borneol x (1 mol / 154.25 g borneol) = 0.001945 mol borneol

mol reactants = mol products

0.169 g camphor x (1 mol / 152.23 g camphor) = 0.00111 mol camphor

Percent yield: 0.00111 mol / 0.001945 x 100% = 57%

Discussion

In this week's lab, camphor was synthesized through reflux. To do so, an alcohol, Borneol, was oxidized to a ketone, camphor, through the use of bleach. The organic layer, dichloromethane, was removed from the bottom of the aqueous layer with a density of 1.33 g/mL. Dichloromethane is the bottom layer because its density is greater than the density of water, which is 1 g/mL. The yield for this experiment was 0.169 g of camphor with a percent yield of 57%. Although this percent yield is not abnormally high, the percent yield could have been higher than it should have due to by-products created by a vigorous runoff reaction that can occur if the bleach is added to quickly. Additionally, if some of the drying agent was left on the watch glass this could also contribute to a higher percent yield.

References

Keller L, Joseph K, and Miguel S. 2023 Fall. CHEM 2081 Organic Chemistry II Lab Manual [lab manual]. Auburn (Al): Auburn University.